

A Biomechanical Comparison Between the Safety-Squat Bar and Traditional Barbell Back Squat

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Abstract

Johansson, DG, Marchetti, PH, Stecyk, SD, and Flanagan, SP. A biomechanical comparison between the safety-squat bar and traditional barbell back squat. *J Strength Cond Res* 38(5): 825–834, 2024—The primary objectives for this investigation were to compare the kinematic and kinetic differences between performing a parallel back squat using a traditional barbell (TB) or a safety-squat bar (SSB). Fifteen healthy, recreationally trained male subjects (23 ± 4 years of age) performed the back squat with a TB and an SSB at 85% of their respective 1 repetition maximum with each barbell while instrumented for biomechanical analysis. Standard inverse dynamics techniques were used to determine joint kinematic and kinetic measures. A 2×3 (exercise $\times$ joint) factorial analysis of variance with repeated measures was used to determine the kinetic and kinematic differences between the squats while using the different barbells. Fisher's least significant difference post hoc comparisons showed that the TB resulted in significantly greater maximum hip flexion angle ($129.33 \pm 11.8^\circ$ vs. $122.11 \pm 12.1^\circ$; $p < 0.001$; $d = 1.80$), peak hip net joint extensor torque ($2.54 \pm 0.4 \text{ Nm} \cdot \text{kg}^{-1}$ vs. $2.40 \pm 0.4 \text{ Nm} \cdot \text{kg}^{-1}$; $p = 0.001$; $d = 1.10$), hip net extensor torque mechanical energy expenditure (MEE; $2.81 \pm 0.5 \text{ Nm} \cdot \text{kg}^{-1}$ vs. $2.58 \pm 0.6 \text{ Nm} \cdot \text{kg}^{-1}$; $p = 0.002$; $d = 0.97$), and ankle net joint plantar flexor torque MEE ($0.32 \pm 0.09 \text{ J} \cdot \text{kg}^{-1}$ vs. $0.28 \pm 0.06 \text{ J} \cdot \text{kg}^{-1}$; $p = 0.029$; $d = 0.63$), while also lifting significantly ($123.17 \pm 20.8 \text{ kg}$ vs. $117.17 \pm 20.8 \text{ kg}$; $p = 0.005$; $d = 0.858$) more weight than the SSB. The SSB resulted in significantly higher maximum knee flexion angles ($116.82 \pm 5.8^\circ$ vs. $115.65 \pm 5.6^\circ$; $p = 0.011$; $d = 0.75$) than the TB, with no significant difference in kinetics at the knee. The TB may be preferred to the SSB for developing the hip extensors and lifting higher maximum loads. The SSB may be advantageous in situations where a more upright posture or a lower load is preferred while creating a similar demand for the knee joint.

Key Words: biomechanics, inverse dynamics, lower extremity, exercise

Introduction

Athletes and recreational weightlifters use the squat to improve their performance, enhance their fitness, and reduce their risk of injury (13,29). Athletes often use the loaded barbell squat as a tool to improve lower-body strength and improve sport performance (13,28). The research supports the squat as an effective exercise to improve jump height, sprint speed, and leg power in a variety of populations (7,8,10,21,31). The squat is also used in rehabilitation settings (11), where it has been shown to be effective in the rehabilitation of anterior cruciate ligament injury (29) and decreasing overall injury risk in athletes (13).

There are many variations of the squat exercise, including high barbell back squat (20), low barbell back squat (20), front squat (28), and safety-squat bar squat (24,26,34). Although many studies have examined the kinematics and kinetics of the squat (see (29) for a review), less is known about squat variations. This is important because variations may be similar to the traditional back squat, or they may place demands on the lower extremity and trunk in unique ways. Knowing these distinctions may be important for program design. For example, one variation may create greater demand at a particular joint or may provide the same demand using a smaller external load.

Directional differences in the kinematics and kinetics of squat variations can be hypothesized with a static stick figure (Appendix A) while keeping in mind the limitations of using such a model. The

static stick figure suggests the following (and their converse): (a) an increased extensor torque about each lower extremity joint with a deeper squat; (b) an increased extensor torque about hip and ankle and a decreased extensor torque about the knee with an increased hip flexion angle; and (c) an increased extensor torque about the knee and decreased extensor torque about the hip and ankle with a more posterior load displacement for a given posture. In addition, the static stick figure also suggests that, for any given posture, an increase in load will result in a proportional increase in all 3 lower extremity joints. For example, doubling the load will double the extensor torques at the hip, knee, and ankle. (Further details concerning the static stick figure and its limitations are provided in Appendix A.)

Despite its limitations, the static stick figure in Appendix A can explain the findings of studies that examined many variations of the squat exercise, such as comparing the high-bar with the low-bar back squat (20,21,36), the front squat with the back squat (11,23,37), restricted with unrestricted squats (18), and the back squat with the box squat (33). However, less is known about the squat using the safety-squat bar (SSB). Although there are a limited number of studies examining the differences between SSB and the traditional barbell (TB), they have all consistently demonstrated that greater loads can be lifted with the TB (24,26,34). Two of these studies examined joint kinematics and found that the SSB resulted in a more upright posture (24,26), and that this may be the most likely explanation for the smaller total load lifted with SSB. However, the picture is less consistent for the other differences between the TB and SSB: Hecker et al. (24) suggest decreased lower extremity muscle activation patterns with the

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SSB, whereas Vantrease et al. (34) suggest similar lower extremity muscle activation patterns with the SSB. In addition, Kristiansen et al. (26) found no difference between exercises for the mean or peak net joint torque (NJT) at the ankle, knee, and hip when comparing an SSB with a TB in the high-bar back squat bar placement (although they did find statistically greater mean knee NJTs with the SSB compared with the low-bar back squat bar placement).

It is important to note that Hecker et al. (24) limited their analysis to joint kinematics and electromyography (EMG), and Vantrease et al. (34) limited their analysis to bar velocity and EMG. Although the EMG signal is a measure of muscle activation, it does not equate to muscle force (30) and may not represent the demand placed on the muscles creating torque about each joint. Therefore, studies using joint kinetic analyses must be conducted to complement those using EMG to further our understanding of the differences in demand between the SSB and TB.

The results of Kristiansen et al. (26) are puzzling. If a greater amount of energy is expended by the system as a whole with the TB (a greater load is moved through the same distance with the TB), then at least one of the NJTs have to be generating and absorbing more energy during the lift. They attempted to explain the differences with the moment arms about each joint (calculated as the distance in the anterior-posterior direction between the ankle, knee, and hip joint centers and the ground reaction force) and found no differences in moment arms for any joint between barbell conditions at 4 different time points during the ascent phase of the squat. Although it is possible that changes in load placement may make compensatory changes in posture that may keep the moment arms about each joint the same (22), these findings are hard to interpret, given the larger loads lifted with the TB. (A larger load with the same moment arms should produce larger joint torques.) One explanation may be that they limited their analyses to 4 time points during the concentric phase of the lift, which may not reflect the energy expended by each NJT throughout the entire lift. Another explanation may be that using static analyses to determine moment arms can lead to erroneous results during dynamic activities (35). A mechanical energy expenditure (MEE) analysis (explained in Appendix B) may provide insights into the differences between the TB and SSB that previous investigations have not.

Therefore, the primary purpose of this study was to compare the demand on the lower-extremity NJTs between the TB and SSB using MEE analyses. As predicted by the static stick figure

developed in Appendix A, we hypothesize that the SSB will require a larger MEE from the knee NJT and a smaller MEE from the ankle and hip NJT compared with the TB. The knee NJT MEE will have a larger contribution, and the hip and ankle NJT MEE will have a smaller contribution, to the total MEE with the SSB. The secondary purpose of this study was to provide mechanical explanations for the findings of the MEE analyses. We further hypothesize a lower load lifted (and total MEE) and a more upright posture with the SSB, which will result from a smaller hip maximum flexion angle (MFA), peak hip NJT, and ankle NJT compared with the TB.

Methods

Experimental Approach to the Problem

Fifteen recreationally trained male subjects performed 8 single-set repetitions of the back squat at 85% of their 1 repetition maximum (1RM) with both a TB and an SSB while instrumented for biomechanical analysis and with a force platform underneath their right foot. Standard inverse dynamics techniques were used to determine maximum joint flexion angles, peak NJTs, and NJT MEE at the ankle, knee, and hip for each barbell condition. Differences between barbell conditions for the kinematic and kinetic variables of interest were examined with a 2×3 (exercise $\times$ joint) factorial analysis of variance (ANOVA) with repeated measures.

Subjects

For recreationally trained lifters, Rhea suggested small effect sizes range from 0.30 to 0.80, whereas moderate effect sizes range from 0.80 to 1.5 (27). Using an effect size of 0.8 (27), an alpha level of 0.05, and a statistical power of 0.8 for a 2-tailed difference between 2 dependent mean values, a sample size of 15 was necessary for this study using G*Power software (version 3.1.9.4) (4,22). Fifteen healthy, recreationally trained male subjects (age 23 ± 4 years, height 177.90 ± 10.18 cm, and body mass 81.64 ± 9.37 kg) who were experienced at performing the barbell back squat participated in this study. Subjects had at least one year of experience performing the back squat with the TB, with an average experience of 60 ± 30 months. Although there was no requirement for previous experience using the SSB, subjects had an average experience of 6 ± 9 months of using the SSB. All subjects were free from injuries that would have limited their ability to



Figure 1. Testing set-up for the traditional barbell (A) and safety-squat barbell (B).

perform the back squat using the 2 different barbells. Before participating in the protocol, which was approved by the California State University, Northridge Standing Advisory Committee for the Protection of Human Subjects, potential subjects were informed of the risks and benefits of the study and provided their written, informed consent.

Procedures

The subjects were asked to refrain from any strenuous lower-body activity for 72 hours before each session and to continue with their normal diet. During the first session, the subjects signed the informed consent form and had their height and body mass measured and recorded. To determine parallel squat depth for each barbell, subjects alternated squatting with an each one, unloaded, until their quadriceps were parallel to the floor. Subjects held that position while the squat rack safety pins were set to the barbell's height. Thereafter, an elastic cord was setup parallel to the safety pins, and then the safety pins were lowered 7.62 cm, so the subjects could touch the cord in the bottom of their squat without touching the safety pins (Figure 1). As a warm-up/familiarization, the subjects performed 4 sets of 10 repetitions with an unloaded barbell (SSB or TB). The barbell used was randomized for each subject but was counterbalanced so that each barbell was initially used by half of the subjects. During the warm-up, the subject's squat stance was marked on the ground, and that same squat stance was used throughout sessions 1–3. After the warm-up/familiarization, the subjects performed a 1RM testing protocol with the initial barbell assigned to start using standard testing protocols. The second session was performed a minimum of 72 hours after the first testing session. The same warm-up and 1RM testing protocols were performed as the first session. The barbell not tested during their first session was tested during their second session.

The third session was conducted a minimum of 72 hours after the second testing session. After an updated body mass was collected, retroreflexive markers were placed on both the left and right side of the body at the following landmarks: femoral greater trochanter, iliac crest directly superior to the greater trochanter marker, lateral and medial femoral epicondyle, lateral and medial malleolus, head of the second and fifth metatarsal, and heel of the shoe. In addition, cluster sets were placed midway of the thighs and shanks that were used to track the segments after the initial calibration trial.

After warming up with the starting barbell (1 × 10 with an empty barbell; 1 × 5 at 50% of 1RM; 1 × 3 at 65% of 1RM; and 1 × 1 at 75% of 1RM), subjects performed 4 sets of 1 repetition with each barbell at 85% (to the nearest 2.5 kg) of 1RM. This was performed twice, for a total of 8 repetitions with each barbell. Subjects squatted using the same foot position (outlined with tape on the floor) and depth from the 1RM testing and a self-selected cadence. They alternated sets between barbells, with the starting barbell randomized for each subject but counterbalanced so that each barbell was the starting barbell for half of the subjects. For example, half the subjects used the following order.

- 4 × 1 SSB at 85% of 1RM (3-minute rest between each set);
- 4 × 1 TB at 85% of 1RM (3-minute rest between each set);
- 4 × 1 SSB at 85% of 1RM (3-minute rest between each set);
- 4 × 1 TB at 85% of 1RM (3-minute rest between each set).

The other half alternated starting with the TB.

Every trial was filmed using a 12-camera motion capture system (Raptor-E, Motion Analysis Corporation, Rohnert Park, CA). The subjects stood on one force platform underneath their right leg

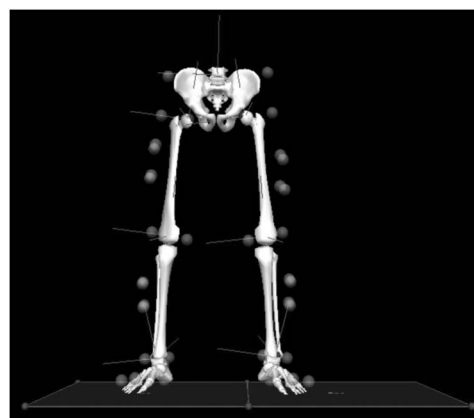


Figure 2. The biomechanical model constructed from the 3-dimensional marker coordinate data. Photo from Visual 3D (C-Motion, Inc, Germantown, MD)

(Kistler, Type 9287BA, Novi, MI). Kinematic data were sampled at 250 Hz. Analog kinetic data were converted to a digital signal of 5,000 Hz, kinematic and kinetic data were synchronized in the motion capture software, and kinematic data were interpolated to match the number of data points of the kinetic data.

Biomechanical Model. Visual 3D (C-Motion, Inc, Germantown, MD) was used to calculate joint kinetics and kinematics for each trial. Marker and force plate data were digitally filtered using a Butterworth low-pass filter with a cutoff frequency of 3 Hz. The marker data were used to construct a model consisting of 7 rigid-body segments: pelvis, 2 thighs, 2 shanks, and 2 feet (Figure 2). Joint angles were calculated as the anatomical angles between 2 segments of the body with the more distal segment as the reference segment (22) and using an X-Y-Z (sagittal-frontal-transverse) Cardan sequence. Maximum flexion angles were the largest values of the flexion angles. Net joint torques were calculated using the same resolution coordinate system as the kinematics and normalized to body mass. The peak NJT (PkJNT) was the largest value of the NJT during the entire repetition. To calculate NJT MEE, the power was first calculated as the dot product of NJT and angular

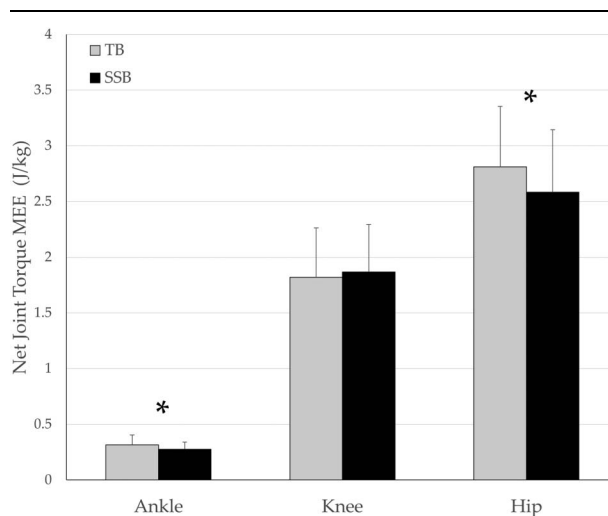


Figure 3. Net joint torque mechanical energy expenditure for the ankle, knee, and hip for the traditional bar and safety-squat bar. (Mean ± SD. TB, traditional barbell; SSB, safety-squat barbell.)

velocity for each joint. Then, MEE was calculated by integrating the absolute power-time curve for each joint during the entire repetition (1). The total MEE was the algebraic sum of the individual MEE (16). The contribution of each individual NJT's MEE to the total MEE expenditure (%MEE), expressed as a percentage, was determined by dividing each MEE by the total MEE and multiplying by 100 (17). All data are reported for the right side.

Statistical Analyses

Before any statistical analysis, the normal distribution of the data was confirmed by the Shapiro-Wilk test ($p > 0.05$). A paired t -test was used to test for differences in 1RM and total MEE between barbell types. A 2×3 (exercise $\times$ joint) factorial ANOVA with repeated measures was used to test for differences in MFAs, PkNJT, and MEE. Greenhouse-Geisser corrections were applied to data that violated sphericity, which was determined using Mauchly's test. Differences in %MEE, which were not normally distributed, were evaluated using Wilcoxon signed-rank test. Post hoc comparisons were conducted using Fisher's least significant difference. The alpha level was set at 0.05. Cohen's d was calculated for effect sizes. Because the subjects fell under the "recreationally trained" category (individuals training consistently from 1 to 5 years), the following interpretations of Cohen's d were used (27): trivial (<0.35); small (0.35 – 0.8); moderate (0.8 – 1.5); and large (>1.50). The 95% confidence intervals (95% CI [lower limit, upper limit]) for differences between bar types are also reported. All statistical analyses were performed using IBM SPSS Statistics for Windows, version 26 (IBM Corp, Armonk, NY).

Results

The mean 1RM loads were 123.17 ± 20.82 kg for the TB and 117.17 ± 20.82 kg for the SSB. Subjects could squat a 4.9% heavier load with the TB compared with the SSB ($p = 0.005$; $d = 0.86$, 95% CI [2.13, 9.87]). This translated into a larger total MEE for the TB ($p = 0.005$; $d = 0.85$, 95% CI [0.08, 0.36]). The TB required 4.65% greater total MEE compared with the SSB (4.95 ± 0.61 J·kg⁻¹ vs. 4.73 ± 0.61 J·kg⁻¹, respectively).

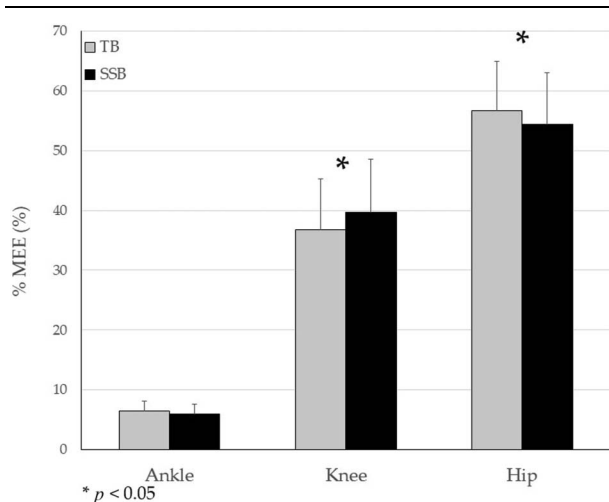


Figure 4. Percentage of each net joint torque mechanical energy expenditure relative to the total mechanical energy expenditure for the traditional bar and safety-squat bar. (Mean $\pm$ SD. TB, traditional barbell; SSB, safety-squat barbell.)

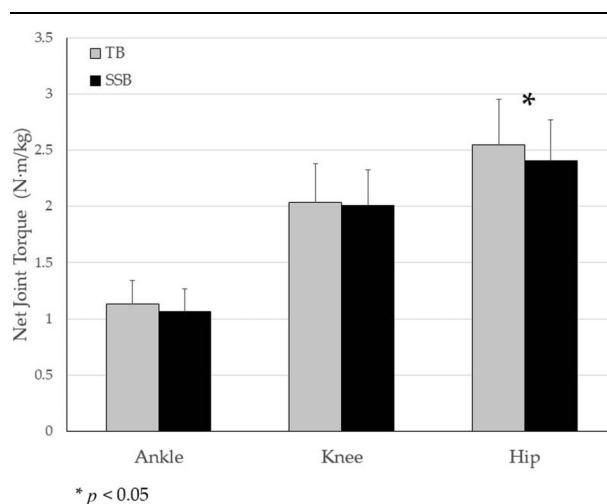


Figure 5. Peak net joint torque for the ankle, knee, and hip for the traditional bar and safety-squat bar. (Mean $\pm$ SD. TB, traditional barbell; SSB, safety-squat barbell.)

For MEE (Figure 3), there was a significant main effect for exercise ($p = 0.005$), a significant main effect for joint ($p < 0.001$), and a significant exercise $\times$ joint interaction ($p = 0.012$). Post hoc analyses reveal there was no significant difference between exercises for the MEE of the knee NJT ($p = 0.268$, 95% CI [−0.13, 0.04]). There was a significant difference between exercises for the MEE of the ankle NJT, with 12.7% more mechanical energy expended by the ankle plantar flexor torque with the TB ($p = 0.029$; $d = 0.63$, 95% CI [0.004, 0.08]), and for the MEE performed by the hip NJT, with 8.1% more mechanical energy expended by the hip extensor torque during the TB ($p = 0.002$; $d = 0.97$, 95% CI [0.10, 0.36]). There was no significant difference in the ankle %MEE ($p = 0.191$, 95% CI [−0.21, 1.24]) between bars, whereas there was a greater knee %MEE with the SSB ($p = 0.011$; $d = 0.85$, 95% CI [0.99, 4.67]) and a greater hip %MEE with the TB ($p = 0.008$; $d = 0.92$, 95% CI [0.92, 3.7]; Figure 4).

For PkNJT (Figure 5), there was a significant main effect for exercise ($p = 0.008$), a significant main effect for joint ($p < 0.001$), and a significant exercise $\times$ joint interaction ($p = 0.011$).

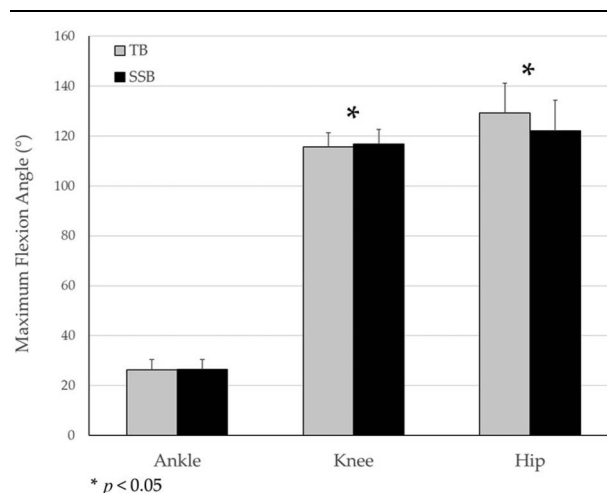


Figure 6. Maximum flexion angles for the ankle, knee, and hip for the traditional bar and safety-squat bar. (Mean $\pm$ SD. TB, traditional barbell; SSB, safety-squat barbell.)

Post hoc analyses reveal there were no significant differences between exercises for ankle PkNJT ($p = 0.066$, 95% CI $[-0.15, 0.01]$) or knee PkNJT ($p = 0.297$, 95% CI $[-0.08, 0.02]$), but there was a significant difference between exercises for hip PkNJT, with the TB requiring a 5.5% greater PkNJT than the SSB ($p = 0.001$; $d = 1.10$, 95% CI $[0.07, 0.21]$).

For MFAs (Figure 6), there was a significant main effect for exercise, a significant main effect for joint, and a significant exercise $\times$ joint interaction (all $p < 0.001$). Post hoc analyses reveal there were no significant differences between exercises for maximum ankle angles ($p = 0.443$, 95% CI $[-0.66, 0.31]$). There were significant differences between exercises for maximum knee angles, with the SSB having a 1% greater flexion angle than the TB ($p = 0.011$; $d = 0.75$, 95% CI $[0.31, 2.03]$), and maximum hip angles, with the TB having a 5.5% greater flexion angle than the SSB ($p < 0.001$; $d = 1.80$, 95% CI $[5.00, 9.3]$).

Discussion

The main purpose of this investigation was to characterize the differences in MEE of the lower extremity NJTs between the TB and SSB. Our results indicate a greater MEE of the hip NJT (moderate effect size) and ankle NJT (small effect size) with the TB and no difference in MEE of the knee NJT between barbells. These findings are in contrast to other studies examining differences in lower extremity demand between the SSB and TB, with one study (24) finding a decrease in demand across the lower extremity with the SSB and 2 others (26,34) finding no difference in demand across the lower extremity between bar types. These findings are not only inconsistent with each other, but they also contradict those predicted by the static stick figure in Appendix A. Possible explanations and interpretations are presented below.

One consistent finding was a lower load lifted with the SSB, although the load lifted with the SSB varied between 89% (24,34) to 97% (26). Our subjects lifted approximately 95% of the TB load with the SSB, which was at the higher end of this range. Greater differences in lower extremity demand between exercises would be expected with a bigger disparity between loads. Although there was only a 5-kg difference in the load used by our subjects, it still resulted in changes in the total MEE of the lower extremity and how the MEE was distributed between the NJTs.

Another consistent finding was that squatting with the SSB resulted in significantly less hip flexion, with small ($<1^\circ$) or non-significant differences in maximum knee and ankle flexion. All studies assessed squat depth by having subjects squat until the thigh was parallel to the ground. With this constraint, it is unsurprising that the knee and ankle flexion angles would be similar for both exercises. If this is the case, then the demand on the lower extremities is determined by the load placement and hip flexion angle (22).

It is possible that the load placement and hip flexion angle interact to keep the center of mass (COM) of the system (barbell plus lifter) centralized within the base of support (22). This may explain why Kristiansen et al. (26) found no differences in moment arms for any joint between barbell conditions at 4 different time points during the ascent phase of the squat. Given a larger load and the same moment arms, the static stick figure would predict larger NJTs across all 3 lower extremity NJTs with the TB. But, they found no significant differences in hip, knee, or ankle NJTs. This disparity in moment arm and NJT findings is hard to explain.

Our results did not match those predicted by the static stick figure (Appendix A), either. There was a significantly larger peak hip NJT with the TB (moderate effect size), with no significant differences

between barbells for the peak ankle or knee NJT. However, our results are closer to the static stick figure when examining MEE: a greater MEE of the hip NJT (moderate effect size) and ankle NJT (small effect size) with the TB and no difference in MEE of the knee NJT between barbells. Mechanical energy expenditure is calculated as the integral of the absolute NJT power-time curve or, equivalently, as the integration of the absolute NJT-angular displacement curve (14). Differences in the NJT at one instant in time may not be significant, but differences in the NJT can create a small effect when integrated over the range of motion of the lift and may be more representative of the demand during the entire lift.

Analyzing the %MEE (the relative contribution) provides another perspective on the differences in mechanical demand between barbells. Differences due solely to the magnitude of the load would lead to proportional differences in the MEE at each NJT, but leave the %MEE of each NJT to the overall NJT invariant, or at least not significantly different between exercises (19). Differences in the %MEE, therefore, suggest a difference in the combined effect of load placement and postures throughout the lift associated with the different bars (Appendix B). Further study, using a variety of loads, would be necessary to determine the consistency of the results across different loading conditions.

The larger MEE and larger %MEE for the hip NJT suggest that difference in MEE was not due solely to larger loads lifted with the TB, but also the load placement and postural adjustments associated with it. The %MEE at the knee was greater for the SSB (moderate effect size), although there was no significant difference in the mechanical energy expended by the knee NJT between exercises. The larger load lifted with the TB offset the greater relative contribution with the SSB. This may also explain why Hecker et al. (24) found smaller EMG activation for the knee extensors with the SSB. Because subjects in their study had a greater disparity between loads (their subjects' SSB load was 89% of the TB, whereas the subjects in this study used approximately 95% of the TB with the SSB), it would seem reasonable that there would be a greater demand at the knee, and hence greater activation of the vastus lateralis, with a greater load on the TB. Given that the ankle NJT MEE was larger with the TB but the %MEE was not, the difference in ankle NJT MEE was more likely due to the increased load with the TB.

Our findings also contribute to the larger body of literature concerning squatting mechanics. A more upright posture during squatting decreases the demand on the hip extensors for the same given load (18,29). Although some studies have found an increased demand on the knee extensors with a more upright posture for the same given load (18,36), others have not (28). However, these relationships are confounded by the fact that more weight can be lifted with a more flexed hip, as demonstrated when comparing the front squat with the back squat (11,23,37) and, to a lesser extent, the high barbell with low barbell squat (20,36). Increases in demand at the knee with a more upright posture may be offset by increases in load with a greater hip flexion angles (23). Therefore, it is important to compare exercises using the same relative, rather than absolute, load.

In addition, our results demonstrate that differences in squat variations cannot be predicted solely from a static stick figure presented in Appendix A, as was performed in some studies (18). Differences in load lifted, placement of the load, and postural adjustments to the load throughout the range of motion must be verified experimentally using inverse dynamics techniques, as was performed in this study. Mechanical energy expenditure analyses may offer additional insights than those provided by NJTs alone because the MEE represents the contribution of that NJT to the energy changes throughout the duration of lift and may be more

sensitive to changes associated with squatting using different barbells. The %MEE may further elucidate these changes that could be hidden by using different loads on the barbells.

Our results suggest that the SSB may be “in-between” the back squat and front squat. The relative intensity using the SSB is higher than performing a front squat with a TB, which has been reported to be about 77–85% of a back squat (11,23,37). The positioning of the SSB would suggest a more flexed posture than with the front squat, although differences in methodologies (i.e., squat depth) make comparisons with other studies (e.g., 37) difficult. Testing all 3 squat variations (front, back, and SSB) under different loading conditions using the same subjects and experimental design would be necessary to make such a determination and is an area of future research.

This study has some limitations that should be considered when interpreting the results. Subjects were more experienced with the TB than the SSB. Differing squat depths (e.g., above or below parallel) and different loads may influence these results (6), and thus, extrapolating them to different depths or relative loads should be performed with caution. Net joint torque analyses assume no biarticular muscles and no cocontraction of uniarticular muscles, both of which may influence the actual muscle demand (9). Although future investigations may address some of these limitations (such as experience, load, and squat depth), others are beyond the scope of current technology (inability to measure muscle forces directly in vivo). However, our results are consistent with findings of previous work on the biomechanical differences between squat variations.

Practical Applications

A greater load could be lifted with the TB. This resulted in a greater demand at the hip extensors and ankle plantar flexors with the TB, with the same demand on the knee extensors, at 85% of a 1RM. The TB would seem to create overall greater demands compared with the SSB. However, the same demand on the knee extensors with less absolute load may be an important consideration in certain training and rehabilitation contexts.

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Appendix A: Static Stick Figure

The static stick figure is the simplest representation of demand on the lower extremity joints and provides a theoretical framework for describing (hypothesizing or explaining) directional differences in squat variations. Directional differences suggest that one variation would require more/less torque at a joint than another, but does not quantify the magnitude of the torque required. The static stick figure represents massless segments that have length but no

width and are connected by frictionless hinge joints. The moment arms of the barbell are assumed to be the only differences creating torque about the joints.

As can be seen in Figure A1, the torque demands on any of the lower extremity joints at any instant are determined by the posture or chain configuration (determined by the joint angles and segment lengths), the placement of the load on the chain, and the magnitude of the load. The combined effects of posture and load placement can best be

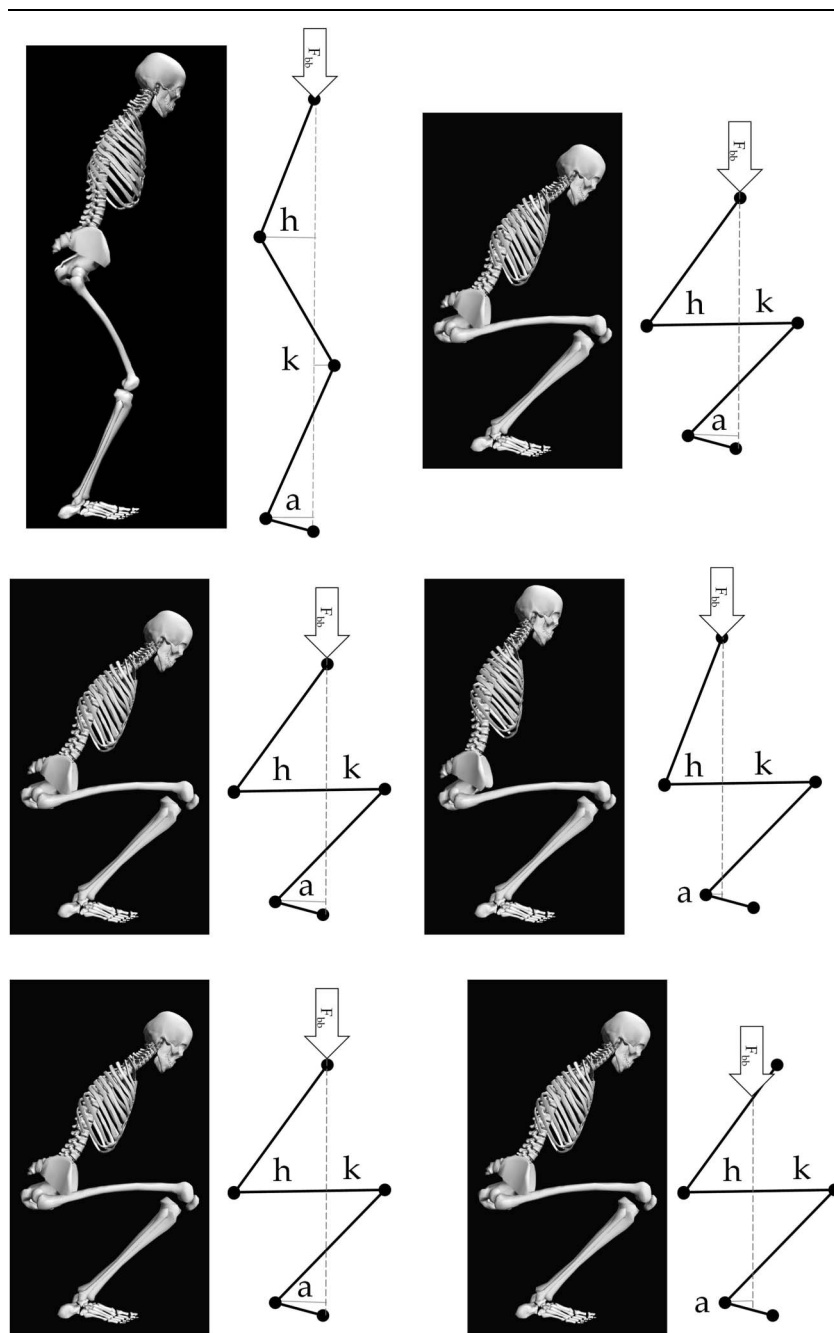


Figure A1. A static stick figure of the squat. F_{bb} = force of the barbell; h = moment arm of F_{bb} about the hip joint center; k = moment arm of F_{bb} about the knee joint center; a = moment arm of F_{bb} about the ankle joint center. Top: greater flexion angles (right) produce larger moment arms about all 3 lower-extremity joints compared with a more upright posture (left). Middle: A smaller trunk inclination angle (right) produces larger moment arms about the knee joint center and smaller moment arms about the hip and ankle joint centers compared with a larger trunk inclination angle (left). Bottom: A lower placement of the load (right) produces larger moment arms about the knee joint center and smaller moment arms about the hip and ankle joint centers compared with a higher load placement (left). Photos obtained using OpenSim (Stanford, CA).

explained by examining the length of the moment arm at that instant, which is the perpendicular distance from the joint center to the force vector created by the weight of the barbell. Figure A1 illustrates how the moment arms at the ankle, knee, and hip change based on changes in squat depth (top), hip flexion (middle), and load placement (bottom). For any given load, an increase in the moment arm will require an increase in torque about that joint, and a decrease in moment arm will require a decrease in joint torque. The static stick figure also suggests that, for any given posture, an increase in load will result in a proportional increase in the demands at all 3 lower extremity joints because the moment arms remain unchanged. For example, doubling the load will double of the torque demands at each joint, provided no postural adjustments occur with the increased load. Because it does not take into account of the mass of the segments or inertial forces, the static stick figure should not be used to calculate actual torque demands. Rather, it is used to predict or explain directional differences (increased or decreased torque demands) by hypothesizing about the changes in moment arms. To determine the usefulness of the static stick figure in generating hypotheses, we compared the results of a static stick figure with a more realistic static model. Both representations used segment lengths as a proportion of subject height as published by Winter (35) for a 1.8-m tall male subject. Mass of the segments and location of the mass centers of each segment were based on de Leva (12) for an 85-kg male subject. Furthermore, a barbell with a mass of 85 kg was added at the level of acromion. The differences in NJTs for 2 postures were compared. Both postures had an ankle dorsiflexion of 40° and a knee flexion of 130°. Posture A had hip flexion angle of 140° (similar to Figure A1, middle left), and posture B had a hip flexion angle of 120° (similar to Figure A1, middle right). The NJT at the ankle, knee, and hip was calculated using the methods of Kane and Levinson (25) using custom code written in MotionGenesis (Motion Genesis LLC, Menlo Park, CA) software. The results are presented in Table A1. Because the static stick figure gave the same directional results (i.e., which NJTs increased and decreased), we feel it is useful in generating hypotheses that can be tested experimentally without the need to resort to more complicated models. However, these hypotheses must be tested experimentally (as was performed in this study) because of the model's limitations. One limitation of the static stick figure is that it does not account for the constraints under which a squat is performed. For example, a mechanical constraint is that the COM of the system (barbell plus lifter) must remain inside the base of support; otherwise, the lifter would fall or have to take a step. Changing the magnitude of the load, placement of the load, or both may require postural adjustments to keep the COM within the base of support (22). These postural adjustments may not be captured with the model. Another limitation is that the static stick figure does not account for the affordances by different postures. The largest torques can be produced by the hip extensors, followed by the knee extensors, and the ankle plantar flexors (2,5). Thus, postures that allow for a greater contribution from the hip extensors can generally support a larger load (11,20,23,36,37). Because the static stick figure only predicts the independent effects of load, load placement, and posture, it needs to be with used with caution when comparing different postures that use different loads (22). Yet another limitation of the static stick figure is that it does not account for the demand placed on a muscle group throughout the exercise; it only provides the torque at one, static instant. It accounts for gravitational, but not inertial, forces. Because the inertial forces and torques are changing throughout the range of

Table A1
Ankle, knee, and hip joint torque for a static stick figure (without segment masses) and an anthropometric model (static stick figure with segment masses) for posture A and B (corresponding to Figure A1, middle left and middle right, respectively).*

		Posture A	Posture B	Difference
Static stick figure				
Torque (Nm)	Ankle	200.74	85.74	-115
	Knee	36.59	151.60	115
	Hip	331.14	216.13	-115
Anthropometric model				
Torque (Nm)	Ankle	264.73	106.20	-158.53
	Knee	194.39	352.93	158.53
	Hip	456.48	297.95	-158.53

*Hip extensor, knee extensor, and ankle plantar flexor torques are positive.

motion, it is more appropriate to use an integral measure, such as NJT impulse or work (3) (Appendix B).

Appendix B: Mechanical Energy Expenditure

The purpose of resistance exercise is to create a progressive overload that will drive neurologic and hypertrophic responses (32). An unresolved issue is which biomechanical measure best represents this overload. Although the net joint torque (NJT) represents the sum of muscular and passive torques necessary to satisfy the equations of motion (38), Andrews (3) suggested that instantaneous measures, such as NJT, are only appropriate as a measure of muscular effort if that measure (NJT) does vary appreciably during the movement. As can be demonstrated by the static stick figure in Appendix A, there are large changes in NJT throughout the range of motion. Therefore, an integrated measure (such as NJT impulse or work) is a more appropriate measure of muscular effort (3). Although one could argue that the PkNJT could be an appropriate proxy for mechanical demand, it is still an instantaneous measure and may not represent the demand throughout the range of motion. For example, consider the 2 hypothetical NJT curves presented in Figure B1 for a range of motion from 0 to 90° (0–1.57 rads). Torque A has a higher peak (1.27 Nm compared with 1

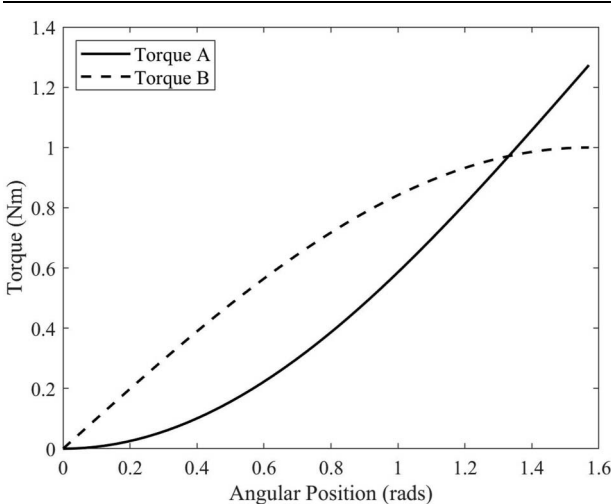


Figure B1. Torque and angular position curves for 2 hypothetical torques. Torque B (dashed line) has a lower peak torque but performed a larger amount of work.

Nm), but torque B has a larger integrated measure or area under the curve (i.e., a larger number of points in the range of motion where the NJT is larger). Torque B represents a work of 1.0 J, whereas torque A represents 0.72 J of work. Although torque A has the higher peak, torque B did more work—which arguably would be a reflection of a greater demand on torque B. Although we agree with Andrews (3) that the integrated measure best represents muscular effort, we believe both should be reported for the reader to draw their own conclusions in cases where one curve has a larger peak, but another has a larger area under the curve.

There are 2 integrated measures for the NJT during a movement. Net joint torque impulse accounts for both the magnitude of the NJT and the length of time the NJT torque is produced, whereas the NJT work accounts for both the magnitude of the NJT and the range of motion over which the NJT is produced (14). There is no consensus as to which measure best represents mechanical demand. However, because Flanagan and Salem (16) found that the sum of NJT work had a higher correlation to the COM energetics during the concentric phase of a squat compared with the sum of the NJT impulse, NJT work would seem to better represent the muscular demand of this activity.

However, as defined by classical mechanics, the NJT work performed during a full repetition of the squat is zero—the negative work during the eccentric phase and the positive work during the concentric phase cancel. Consider the angular position, angular velocity, and torque for a hypothetical joint motion presented in Figure B2. The joint moves through 90° (1.57 rads), pauses, and then returns back to the starting position (0°). The torque's power and energy of the segment are presented in Figure B3. Positive power indicates a concentric action, whereas negative power indicates an eccentric action. The torque first put energy into the segment from 0 to 90° and then, after the pause, removed energy from it from 90 to 0°. The resulting change in energy was 0 J. As work (and the change of energy) is the area under the power-time

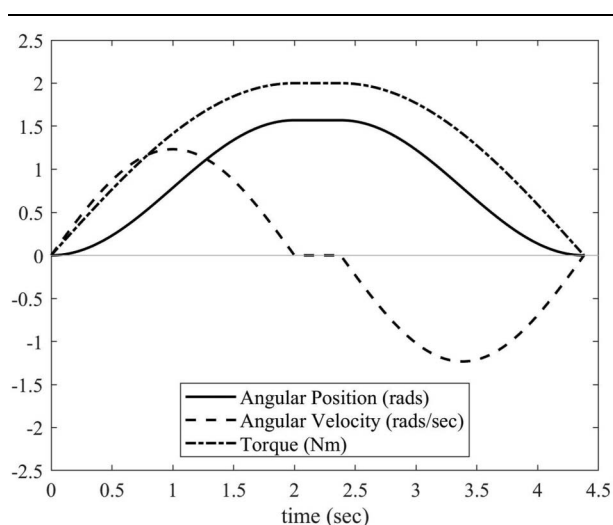


Figure B2. Angular position (solid line), angular velocity (dashed line), and torque (dotted dashed line) for a hypothetical joint movement.

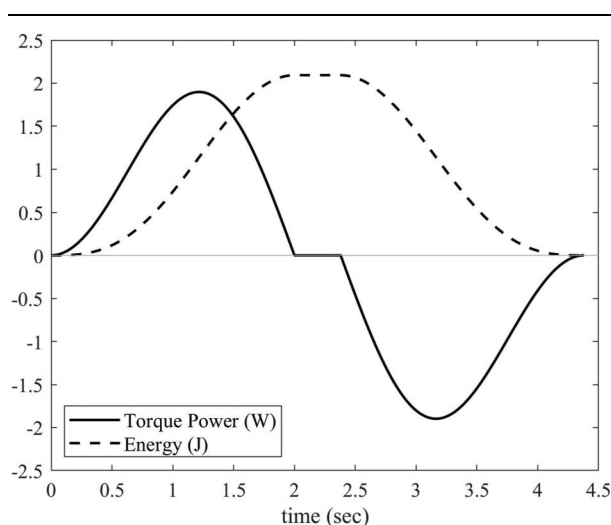


Figure B3. Torque power (solid line) and energy of the segment (dashed line) for a hypothetical joint movement.

curve (14), it follows that the positive and negative areas must cancel because the change of energy was zero (Figure B4, top). Aleshinsky (1) suggested summing the positive work with the absolute value of the negative work in these situations (see bottom of Figure B4), and referring to this summed value as mechanical energy expenditure (MEE), sometimes also referred to as “absolute work.” This would seem to be a more satisfying measure of muscular demand during an activity. An additional benefit of using MEE as a dependent variable is that the MEE of several joint torques can be summed together, representing an overall MEE. This “allows estimating MEE for motion in an uncontroversial way” (38). Then, each individual NJT MEE can be expressed as percentage of the total lower extremity MEE (%MEE). Changes due solely to an increase in load would change the MEE of each joint, but the %MEE would remain the same (19). For a hypothetical example, see Table B1. Condition A represents a baseline case. Condition B1 has the activity performed in an identical manner, except the load was doubled. Note that this doubled the MEE of each NJT, but each NJT %MEE remained unchanged. Changes in the %MEE would reflect changes because of load placement and postural adjustments. Referring back to Table B1, the load, and MEE, was once again doubled, but the %MEE changed for condition B2. Something in addition to load was responsible for the change in %MEE. However, one would not be able to ascertain this just by noting that the individual NJT MEE was larger. For an experimental example, one study examining the effect of 3 different step exercises (step-up, lateral step-up, and step-down) using the same load on a barbell found that the total MEE was not significantly different between step conditions, but the %MEE for the ankle, knee, and hip NJT was significantly different (15). Because the load and step height were the same across conditions, the authors concluded that the differences in %MEE were due to postural adjustments necessary to complete the exercise. For these reasons, we believe MEE analyses are the most appropriate to use when examining squats performed with different types of bars.

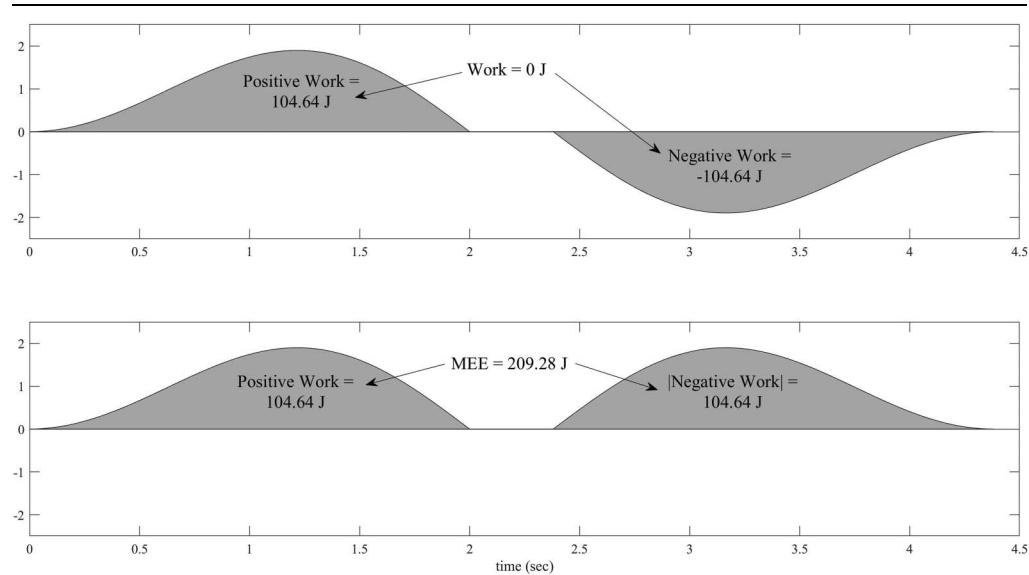


Figure B4. The difference between work (top) and mechanical energy expenditure (MEE; bottom) of the joint torque for a hypothetical joint movement.

Table B1						
Differences in MEE and percent of total MEE (%MEE) for 3 different conditions.*						
	Condition A		Condition B1		Condition B2	
	MEE (J)	%MEE (%)	MEE (J)	%MEE (%)	MEE (J)	%MEE (%)
Ankle NJT	50	10	100	10	80	8
Knee NJT	200	40	400	40	440	44
Hip NJT	250	50	500	50	480	48
Total	500	100	1,000	100	1,000	100

*NJT = net joint torque; MEE = mechanical energy expenditure.